OUR PLANETARY CONDITION

Foundations for a Politics with the Earth

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List of Terms

Agency & Agents

Agency refers to the capacity of entities—whether nonliving, living, or technological—to influence the world around them. In traditional human-centered frameworks, *agency* is typically understood as the intentional and conscious actions of *agents*, with accountability tied to specific processes, goals, and outcomes. This perspective assumes that actions—modeled on human behavior—stem from intentions, are linked to mental states, and aim toward specific objectives. However, such a view is inadequate for understanding the vast, interconnected dynamics of *nonhuman* entities, which often lack or cannot express conscious intentions or deliberate objectives according to human models. To address these parameters, the humanities and social sciences increasingly conceptualize *agency* as relational, arising through processes of interconnected world- and meaning-making. In this expanded framework, *agents* are not restricted to humans but include any entity capable of introducing changes or variations. From geological processes like volcanic eruptions to the behaviors of living organisms or the operations of technological systems, *agency* manifests in diverse and often overlapping forms. Accordingly, we describe *planetary agency* as the manifestation of any form of planet-shaping realized by various *planetary agents* within, on, throughout, and around the Earth. We thus recognize all planetary entities—whether nonliving, living, or technological—as political agents that influence our planetary condition.

Agency of Life

The *agency of life* is the capacity of living organisms to exert influence at diverse spatiotemporal scales. We utilize this concept to emphasize the dynamic, emergent, and often unintentional processes through which living entities—from microbes to complex organisms—engage with and transform their planetary surroundings via growth, adaptation, symbiosis, and other forms of *interaction* and *intra-action*. An example of *interaction*, highlighting the autonomy of entities, is microbial decomposition, where soil bacteria engage with dead organic matter, transforming the latter into nutrients for plants. Conversely, an example of *intra-action*, showcasing the intertwined nature of entities, is a coral reef ecosystem, in which coral polyps and their symbiotic algae co-constitute a living system in which the former's structure and the latter's photosynthesis are inseparable processes. The *agency of life* highlights the correlation and permeable boundary between biological and non-biological that persistently and significantly shape our planetary condition.

Agency of Nonlife

The *agency of nonlife* is the capacity of nonliving entities to exert influence across spatiotemporal scales. We utilize this concept to challenge the notion that nonliving entities—from tectonic shifts to hydrological cycles and atmospheric dynamics—are passive or inert; instead, it emphasizes their active role in shaping the Earth via physical, chemical, and mechanical processes of *interaction* and *intra-action*. An example of *interaction* is tectonic activity, where geodynamic processes actively and continuously transform landscapes, influencing erosion and carbon cycling, ultimately affecting planetary climate over geological timescales. Conversely, Earth's water cycle illustrates the *intra-action* of nonliving agencies, as evaporation, precipitation, and runoff co-constitute a system that regulates global weather patterns and sustains ecosystem functions. The *agency of nonlife* highlights the correlation and permeable boundary between nonliving and living entities, recognizing the formative role of these dynamic processes in determining our planetary condition.

Agency of Technology

The *agency of technology* is the capacity of technological entities to exert influence at spatiotemporal dynamics across human and nonhuman domains. We utilize this concept to emphasize the dynamic, emergent, and often unintended (sometimes autonomous) processes by which technological entities—such as machines, algorithms, networks, and infrastructures—engage with and reshape the planet by mediating energy, information, and material flows through *interaction* or *intra-action* with other agencies. An example of *interaction*, highlighting a distinct and autonomous aspect of this concept, is human energy infrastructure such as hydroelectric dams, wind and solar farms, and nuclear power plants that interact with nonliving planetary agencies to generate energy, alter ecological systems, and affect social dynamics. Conversely, the global internet exemplifies *intra-action*, demonstrating the entangled nature of technological systems, where hardware, software, and human engagement co-constitute a distributed system that blurs the boundaries between human and technological agency, facilitating communication, economies, and cultural practices. The *agency of technology* acts as a mediator between planetary agents and agencies, emphasizing the co-evolution of human and nonhuman systems that increasingly contour our planetary condition.

More-than-Human

The concept of the *more-than-human* refers to the notion of the world as an intricate web where humans exist alongside countless other entities with capacities. The human body comprises countless more-than-human beings (from bacteria to microplastics). The *more-than-human* signifier is thus often used correspondingly with the *nonhuman* to describe entities *other* than humans. The aim is to foreground the agency of nonhuman beings or entities and challenge the inclination of the *nonhuman* signifier to define them negatively by highlighting that they are 'not human.'

Nonhuman

The *nonhuman* signifier describes all life forms or entities that are *not* human, including natural phenomena. It highlights that humans are part of the material realm, which humanist discourse frequently designates as the 'animal' or 'natural' world. References to *human* and *nonhuman* beings or animals aim to challenge categorical distinctions of humans as separate from all other animal species and life forms; a further relativization of the *human* category is the separation of *human* and *other* animals. The *posthuman* category is related to the *nonhuman* but includes human beings, stressing the constructed nature of the *human* signifier.

1 Introduction: Emerging Awareness of Our Planetary Condition

From medical to earth system sciences, from planetary health (Whitmee et al., 2015) to planetary emergency (Lenton et al., 2019), the natural sciences have for some time now approached the Earth as a planet to be “made whole” (Dry, 2019). If belatedly, the humanities and social sciences, too, progressively move the *planetary* into their disciplinary focus. From planetary justice (Lobo et al., 2024; Pedersen et al., 2024) to planetary law (French & J. Kotzé, 2021; Lawrence, 2018) to planetary histories (Bashford et al., 2023; Chakrabarty, 2021), scholars have begun to engage deeper with the implications derived from knowledge of how planets work. Hence, what happens if the Earth is no longer primarily perceived as a metabolic system but as an agential complex, a planet?

In this book, we show that the notion of the planet in general and our respective planetary condition, in particular, allow for a broader perspective that ultimately changes not only the universal understanding of the human condition but also how politics, in general, is perceived, analyzed, and implemented. This study aims to elucidate the planetary agencies that co-constitute *our* planetary condition and identify the demands that arise thereof for a respective politics *with* the Earth.

We do not ‘live on’ but are ‘part of’ a planet. As planet Earth is ever-changing, so are the relations societies create with it. This has brought into focus a distinct set of phenomena in which planetary and anthropogenic forces intersect, ranging from (induced) seismicity to (anthropogenic) space weather. Mutually complementary to the planetary sciences, the planetary humanities and social sciences can offer frameworks for understanding how societies can engage with planetary forces, some of which are beyond human control. These disciplines and fields can help us navigate complex decisions about which practices to sustain, transform, or relinquish in the face of planetary challenges. In other words, understanding how planets function—

geophysically, ecologically, and socially—is essential to rethinking societies and their political strategies planetarily.

To push against the latent discomfort that the ‘planetary’ signifier may stir for the universalizing or assimilatory connotations associated with it (cf. Ferdinand, 2022; Jazeel, 2011; Kennedy, 1999; Mouffe, 2005, p. 103), we deliberately speak of *our* planetary condition as indefinite, the only one that ‘we’ as co-authors may describe from our necessarily human-centered but self-aware, limited perspective, given that we represent only a minor fraction within the planetary subset of human beings. Whereas the planetary condition connects all Earthly beings as collectively responsible subjects, it bears highly differentiated implications for the innumerably diverse worlds it refers to (Gabrys, 2018). The first-person plural possessive may be “problematically presumptive,” carrying the bitter taste of exclusion or othering (cf. Tsing et al., 2024, pp. 12-13). However, in the scaffolds for a planetary politics charted in this work, ‘our’ aims to materialize simultaneously as an inclusive term that seeks to identify, approximate, and weave shared ideals, objectives, and conflicts of various planetary agencies toward a planetary polis that transcends individual differences.

In this first chapter, we draw on the interrelated concepts of *thought collectives* and *thought styles* to examine who or what plays a role in shaping our planetary condition and how we conceive of these agencies, deriving our analytical scheme from their essential characteristics. Chapters 2, 3, and 4 analyze three distinct types of planetary agencies that co-constitute our planetary condition on Earth. Building on this foundation, Chapter 5 develops a framework for a planetary politics that mobilizes a diverse assembly of agents and agencies. The concluding sixth chapter synthesizes insights gained from the analyses of planetary agencies, envisions the future of planetary politics, and provides a case study on more-than-human decision-making.

1.1. Planetary Thought Collectives

In his 1935 book *Genesis and Development of a Scientific Fact,* science theorist Ludwik Fleck introduced the interlinked concepts of *thought collectives* and *thought styles* to highlight the collective and relational nature of scientific discovery, emphasizing how knowledge emerges from the interplay between researchers, the phenomena they study, and the broader epistemic frameworks that shape their inquiries (Fleck, 1979, p. 27). A *thought collective* is a community of researchers who produce or develop knowledge through shared cultural customs, methodologies, and practices, forming a foundation for collective understanding. This shared *thought style* provides researchers with common concepts and practices, fostering collaboration and creating a shared language. However, it can also shape and limit the scope of scientific inquiry, as the overarching thought structure influences what is considered meaningful or valid. While different subgroups may develop distinct concepts, theories, and models within any given thought style, they constitute an overarching democratic community that pursues knowledge production as a collective process.

In planetary research, the interdisciplinary and transdisciplinary thought collective focused on the Earth exemplifies how diverse disciplinary perspectives converge to form a shared framework—or thought style—for addressing complex planetary challenges. Recognizing the interplay of different subgroups and perspectives within this collective allows us to improve our understanding of how knowledge is constructed and how it may evolve to address the complexities of our planetary condition.

While the natural sciences have persistently studied our planetary condition, recent efforts by several individuals and organizations—many rooted in humanities and social sciences approaches—have explicitly dedicated themselves to defining and addressing the complexities of this condition (see www.planetaryportal.org). These initiatives mark the founding of a genuine 'planetary' thought collective, an epistemic network of interconnected ideas and practices whose intellectual origins span centuries, if not millennia. To better understand contemporary planetary thought styles, we first turn to three historical forerunners who approached the planetary condition from distinct perspectives. These examples highlight the pluralistic nature of planetary thinking, demonstrating how diverse communities across different times and locations have grappled with planetary questions, laying the groundwork for our efforts.

Firstly, many Indigenous groups developed and now practice planetary knowledge that never conceived of the world as a dichotomous formation of nature and culture or practice and theory (Cusicanqui, 2012; Sundberg, 2014; Todd, 2016), as particularly feminist contributions to the debate point out (cf. Irni, 2013; TallBear, 2017; Willey, 2016). Indigenous cosmologies, in all their breadth and diversity, are more than archives of practices of thought, faith, and action; they are relevant 'situated knowledges' which remain closely tied to the practices of those who produce it (Nakashima et al., 2012; Whyte, 2013; Wildcat, 2009). Contrasting such planetary cosmologies with modernity's dominant notion of the global, Gayatri Chakravorty Spivak has proposed for "the planet to overwrite the globe" and to turn to "[p]lanet-thought" (Spivak, 2003, p. 72). Retiring our role as "global agents," she holds, we should "imagine ourselves as planetary subjects" (Spivak, 2003, p. 73):

> Globalization is the imposition of the same system of exchange everywhere. In the gridwork of electronic capital, we achieve that abstract ball covered in latitudes and longitudes, cut by virtual lines, once the equator and the tropics and so on, now drawn by the requirements of Geographical Information Systems. […] The globe is on our computers. No one lives there. It allows us to think that we can aim to control it. The planet is in the species of alterity, belonging to another system; and yet we inhabit it, on loan. It is not really amenable to a neat contrast with the globe. I cannot say "the planet, on the other hand." When I invoke the planet, I think of the effort required to figure the (im)possibility of this underived intuition. (Spivak, 2003, p. 72)

Secondly, Alexander von Humboldt, after engaging with Indigenous communities in South America and linking their knowledges to European scientific traditions, outlined in an 1834 letter the vision for what would become his final and most remarkable scholarly work, the five-volume *Cosmos*:

> The mad fancy has seized me of representing in a single work the whole material world,—all that is known to us of the phenomena of heavenly space and terrestrial life, from the nebulae of stars to the geographical distribution of mosses on granite rocks, and this in a work in which a lively style shall at once interest and charm. Each great and important principle, wherever it appears to lurk, is to be mentioned in connection with facts. It must represent an epoch in the mental development of man as regards his knowledge of nature. […] The whole is not what has hitherto been commonly called 'Physical Description of the Earth,' as it comprises all created things—Heaven and Earth. (Humboldt et al., 1860, pp. 15-17)

In his *Cosmos*, Humboldt hoped to present an imaginative synthesis of all knowledges about the world, visualized by captivating maps and drawings. Most remarkable from a planetary perspective is that the polymath devoted *Cosmos*' third volume entirely to "'cosmical phenomena," ranging from the stars and planets to the velocity of light and comets" (Wulf, 2015, p. 269). Even if some of its findings are, of course, by now outdated, Humboldt's magnum opus remains a significant milestone in the history of science and his methodology of "thinking observation [*denkende Betrachtung*]" as "a meaningful ordering of the appearances of nature … [that is] deeply penetrated with the belief in an ancient internal necessity" (Humboldt, 1845, pp. 31-32) could not be more topical for our planetary condition.

Thirdly, Astronauts of the first Space Age supplemented the “view from above.” This is how the pilot of the Apollo 14 lunar mission (and the sixth human to set foot on the Moon) described his experience of looking at Earth from outer space:

> You develop an instant global consciousness, a people orientation, an intense dissatisfaction with the state of the world, and a compulsion to do something about it. From out there on the moon, international politics looks so petty. You want to grab a politician by the scruff of the neck and drag him a quarter of a million miles out and say, “Look at that, you son of a bitch.” (Mitchell, 1974).

This so-called *overview effect* is, in turn, surpassed by what has recently been dubbed the *ultraview effect*, i.e., “the view of space seen from outside the Earth’s atmosphere, either from Earth orbit or lunar orbit” (Weibel, 2020, p. 5). Nevertheless, the inverse perspective, too, increasingly affects citizens in thoroughly urbanized metropoles, their cosmic relations and realities manifesting skewed and detached from their planetary home. During a blackout following the Northridge earthquake in 1994, for example, Los Angeles emergency services were flooded with calls from citizens concerned about the “giant silvery cloud” hovering over the city (Sharkey, 2008). Usually obscured by light pollution, the Milky Way’s visible presence had spawned fear among Los Angelinos.

Such views of the Earth encapsulate the essence of the planetary thought collective: we are, always were, and will be in space. As planetary creatures aboard ‘Spaceship Earth,’ equipped with finite resources, we hurtle through space at immense speed (Fuller, 1969). The first photographs of Earth not only inspired James Lovelock and Lynn Margulis’ Gaia hypothesis in the 1970s but also catalyzed green social movements (Arendt, 1961; DeLoughrey, 2014; Poole, 2010) and reshaped planetary epistemology. This transformation shifted the focus from conquering the skies or gaining military advantages to fostering peaceful cooperation in solar system exploration and safeguarding the planet—reflecting shared concerns about nuclear self-annihilation and humanity’s collective habitat. The renaissance of space exploration, ranging from Solar Orbiter to Perseverance, continues to offer lessons from interplanetary comparisons for a recursive understanding of Earth(lings) (Lapôtre et al., 2020). This perspective challenges the “container ideologies” of “methodological nationalism” (Beck & Grande, 2010), problematizes identitarian exclusions, and dismantles the nature-culture dichotomy. It also humbles human hubris: no more ‘crown of creation,’ no more Promethean mastery of productive forces.

Figure 1 (above): Overview Effect. Astronaut Tracy Caldwell Dyson looks through a window in the Cupola of the International Space Station (ISS). NASA/Tracy Caldwell Dyson. **Figure 2** (below): Ultraview Effect. NASA/M. Justin Wilkinson, Texas State U., Jacobs Contract at NASA-JSC and Mark Matney, NASA-JSC.

1.2. Planetary Thought Styles

The ‘planetary’ has emerged as a novel category of the political alongside the hitherto dominant global, which can be found in concepts ranging from globalization to global governance. Peculiarities of the planetary begin with its etymology: the globe as “the ball” (from Latin *globus*) stands motionless against the planet as “the wanderer” (from Greek *planētēs*, ancient Greek *πλάνητες ἀστέρες*). Merriam-Webster’s dictionary defines the globe as the “spherical representation of the Earth” and the planet as “a body revolving around a star and this planet Earth.” One may say Earth is one planet among others and thus comparable to these others. It is the interplanetary comparison that allows careful consideration of the different qualities a planet develops over time, expanding them as “analogues, experiments and archives” for Earth (Lapôtre et al., 2020), such as seismic events on Mars (‘marsquake’) and on Earth (‘earthquake’) (Witze, 2022). “The planetary is a new ‘cosmology’—emerging as an alternative to the social, cultural and political-economic orders of global modernity,” remarks astrobiologist Adam Frank fittingly; “[t]he planetary is a radically new worldview and paradigm grounded in revolutionary scientific advances,” from geospheres to biospheres to technospheres, and “promises a new design for our future in a climate-changing world” (Frank, 2024).

Understanding our planetary condition and thus thinking in planetary terms first and foremost means recognizing the Earth as a planet—epistemologically, ontologically, and ethically. Historically, one can roughly distinguish between three overlapping phases in the twentieth and twenty-first centuries. In the first phase, the Cartesian divide between nature and culture was epistemologically constitutive for the ontological recognition of a (global) environment related to the normative goal of protection, resulting in conservation practices such as national parks. In the second phase, particularly in the Earth System Sciences, a system perspective was developed that allowed us to see the Earth as a complex system with the normative anchor point of sustainability. This perspective was translated into concepts like sustainable development in the Brundtland Report and applied in political arenas in conventions such as the UNFCCC. As we show in this book, relationality, as the core epistemological tool, has most recently gained application in demonstrating that we are part of a planet whose habitability is central to an emerging planetary politics that genuinely recognizes and includes diverse planetary agents and agencies.

cartesian divide	system perspective	relationality
(global) environment	Earth (system)	planet
protection	sustainability	habitability

Table 1: Phases of Recognition—Earth as a Planet

Such a planetary perspective means acknowledging that human (co-)existence is inescapably tied to a planet that keeps metamorphosing and expanding—spatially from Earth's core to interplanetary space, temporally from nanoseconds to cosmic time, and materially from elementary particles to the dark matter of the universe. Sociologists Nigel Clark and Bronislaw Szerszynski characterize the different phases of this ever-changing planet as a "planetary multiplicity"—and societal relationships to it as "earthly multitudes" (Clark & Szerszynski, 2021). Questions driving a broad research program informed by our planetary condition include:

- Retrospective: How have planetary agencies shaped human societies throughout history? In what ways have diverse societies cultivated capacities to transform Earth's systems, and how have these capacities been unevenly distributed or contested?
- Contemporary: How can societies today navigate the challenges of irregular yet recurring planet-shaping agencies beyond their control? What does it mean to release or engage with such agencies responsibly? Under what circumstances should these agencies be untouched, restrained, or harnessed?

- Prospective: What planetary agencies beyond human influence might emerge or intensify in the future? Which forces might we strategically align or collaborate with, and under what terms? Conversely, which alliances with planetary forces should we seek to dissolve or avoid—if such choices remain possible?

We can identify typical cases where planetary and anthropogenic agencies mutually intersect and create our planetary condition throughout human history and across the planet. To provide but a brief survey, these intersections include, for example, (I) the earthquake in Lisbon in 1755 (resulting in political turmoil) and induced seismicity (through artificial vibrations of dams); (II) the volcanic eruption of Tambora in 1815 (causing famines) and human influences on volcanic activity (through melting ice shields); (III) solar storms in the USA in 1967 (almost leading to a third World War) and anthropogenic space weather (courtesy of nuclear bombing); (IV) the Bhola cyclone in 1970 (facilitating the foundation of the state of Bangladesh) and human-engendered cyclones (because of anthropogenic climate change).

When it comes to an understanding of our planetary condition, this has resulted primarily in the formulation of a range of theoretical approaches to overcome dualisms, including a posthumanism that emphasizes nonhuman agency (Tsing, 2015), a new materialism that proposes categories such as *hyperobjects* to capture that which can hardly be sensed but is everywhere (Fox & Alldred, 2015; Harman, 2018; Morton, 2013), or relational (Walsh et al., 2021) neologisms like *natureculture* (Descola, 2013), *spacetimemattering* (Barad, 2007), or *technogeographies* (Gabrys, 2016).

Overall, we identify five perspectives in planet-thought: metabolic, recentering, transversal, interrelational, and normative. Our planetary condition may be understood theoretically from a *metabolic perspective* concerning the flow of matter between the planet and humanity—without equating the two spheres or lapsing into material relativism. It can be secondly understood from a *recentering perspective* that detaches humans from their position of uniqueness—without releasing them from responsibility. Thirdly, it can be interpreted from a *transversal* perspective because it connects things and concepts such as nature and culture—without crucially merging them. Fourthly, an *interrelational perspective* allows us to see the interactions between various planetary agents. Lastly, a normative perspective enables us to evaluate whether the ways in which these planetary agencies are at play are desirable, particularly in terms of the habitability of this one planet, Earth.

Applying these various perspectives, the challenge is to account for the multitude of views of the planetary, which do not, conversely, lead to a multitude of worlds but to *one* world, a world that can be described in various ways and where extremely diverse behaviors are possible. Planetary knowledge neither relativistically aims at letting everyone live in their specific world nor monastically at creating a unified science or a dualism of scientific and non-scientific approaches to the world. It is pluralistic, or rather *pluriversalist* (Mitchell, 2008; Reiter, 2018). Planetary thinking conceives the world as variously entangled, corresponding to a non-linear idea of the causality of complex interdependencies and a multi-level system of inter- and transdisciplinary methods. The design of planetary ecologies of knowledge is an evidence-based activity in search of variability and permanently constituting shapes. This design activity embodies the genuine human aspect of planetary knowledge. The shaping of knowledge is an act carried out not merely intellectually but as part of a comprehensive conduct of life. Planetary knowledge is reflective or has a "diffractive" quality (Barad, 2007, passim). Ecologies of knowledge (Akera, 2007; Rahder, 2020), namely, an "ethico-onto-epistemic" frame (Barad 2007, passim) that is capable of integrating a significant number of assumptions and explanations about the Earth as a planet, changes the more we know. The thinking and non-thinking parts of the planet thus converge.

This diffracts our conception of how said parts actively co-shape this world and what these *planetary agencies* imply for a politics practiced *with* the Earth. We understand any forms of planet-shaping as manifestations of planetary agency. These entail the relational and emergent dynamics that *all* entities within and surrounding the Earth unleash, regardless of whether they follow cognitive processes, intentions, conscious actions, individual or self-aware motives, and irrespective of whether we classify them individually as 'intelligent' in the conventional sense or collectively as a "planetary-scale process" (Frank et al., 2022). Instead of as "an innate, restrictive set of behaviors," James Bridle defines intelligence as "something which arises from interrelationships, from thinking and working together […]. If all intelligence is ecological—that is, entangled, relational, and of the world—then artificial intelligence provides a very real way for us to come to terms with all the other intelligences which populate and manifest through the planet" (Bridle, 2022, p. 57).

A comprehensive discourse on nonhuman agency has exposed these and other anthropocentric markers as imprecise. Their application in political theory has caused political instruments to be unable to answer to the interests and needs of planetary subjects whom humans readily and wrongfully discard as objects (cf. Bridle, 2022, p. 18; Meijer, 2019a, pp. 3-5): *More-than-human agents* (Abram, 1996; Bridle, 2022), *non-human actants* (Latour, 2005), *companion species*, or *(political) animal agents* (Bovenkerk & Meijer, 2021; Donaldson & Kymlicka, 2011; Haraway, 2003), *intra-agents* (Barad, 2007), *vital materials* (Bennett, 2010), *posthuman agents* (Braidotti, 2020; Braidotti & Jones, 2023), *plant agents* (Gilroy & Trewavas, 2023), *artificial agents* (Bostrom, 2005), and the agentic assemblages, networks, confederations, and other sorts of constellations they form belong to an unwieldy crowd of *planetary agents* and *agencies*. Their actions may further take shape as thing-power, intra-actions, interactions, vitality, physical forces or phenomena, consumption, reproductive or competitive practices, expressions of preferences, beauty, desires, intentions, etc. Any effects that materials, living organisms, technological entities, weather phenomena, or ecosystems cause and affect other agents and agencies with, hence, co-constitute our planetary condition alongside human zones of influence and thus implicitly or explicitly participate in a planetary politics (Barad, 2007, pp. 178, 214; Bennett, 2010, pp. xvi, 51, 121; Blattner et al., 2020; Bovenkerk & Meijer, 2021, p. 54; Coole & Frost, 2010, p. 9; Fratzl & Schäffner, 2021, p. 38).

Not attempting to unify the many diverse groups or kinds of agents and agencies into one, the ‘planetary’ signifier instead conveys what they have in common. As all of them partake in dynamics that bear weight on or lift the weight off the Earth’s crucial biophysical systems, none are excluded from the circle of collectively responsible planetary subjects (cf. Gabrys, 2018). Albeit to drastically unequal degrees (Yusoff, 2018), human agents are distinctly accountable for the phenomena that gradually push the Earth toward potential ecological collapse (cf. Asenbaum et al., 2023; Cudworth & Hobden, 2018). However, as none of the many planetary agents, not even human ones, can exist or act independently, human politics can no longer ignore the electrons, molecules, microorganisms, atmospheric and oceanic currents, fungi, insects, animals, and fundamental forces they implicitly or explicitly ‘collaborate’ with. To support an overall flourishing of life and build a democratic, ‘more-than-human place,’ decision-making requires input from all contributing agents (cf. Blake & Gilman, 2024, pp. 95-96; Bridle, 2022, p. 207). Hence, in contrast to flat ontological (Harman, 2018), transhumanist (Aydin, 2017; Baillie & Casey, 2005; Bauer, 2010), or ecocentric thinking (Leopold, 1949), a planetary view discloses a vastly heterogeneous political arena, where humans, while in charge, no longer pose the absolute authority. Instead, a hardly manageable crowd of more-than-human political actors—human ones included—are set to negotiate the terms of living together in manners that depend on what or, much rather, 'who' they are—their ontological types (Bennett, 2010, p. 9)—and in directions that keep the planet habitable (Bignall, 2023; Braidotti, 2020). To access their crucial input, a planetary politics seeks to secure, restore, and politically navigate the conditions planetary agents require to realize their agencies (cf. Bignall, 2023, p. 105). Before conceptualizing institutional structures and practices, we thus need to get to know these relational, planet-shaping agencies.

We no longer ask which living, nonliving, or technological entities can or cannot be considered agents of interest in politics. Neither do we ask whether these agents deserve rights analogous to human rights. To serve habitability and multispecies flourishing, political institutions need to access the many voices of the planet more directly than is possible via human proxies, even if these draw on expert knowledge (cf. Blake & Gilman, 2024, p. 177). *Planetary institutions* thus not only need to operate at multiple scales, beyond nation states, and under functionally resultant authority from various directions; the *planetary sapience* (Bratton, 2021) on which their decision-makers draw must emerge as directly as can be from the planet's biogeophysical realities (Blake & Gilman, 2024, pp. 176, 109, 125, 164-165). The task is to pave the way for integrating planetary sapience into politics. While, for example, theoretical perspectives of new materialist, posthumanist, feminist, animal and plant theory, and more-than-human rights discourses have developed insightful perspectives, it appears to us that they have barely grappled with the practical implications of planetary agencies. Both their roles in co-constituting our planetary condition and their consequences for a planetary politics are significant. One might argue that the theoretical debate has stalled in the last decade, with relevant political institutions and processes barely evolving.

Consequently, building on what Jennifer Gabrys terms a *planetary praxis* (Gabrys, 2018), we approach the agencies shaping our planetary condition through an inductive lens, striving to engage them with minimal theoretical preconceptions. We ask how diverse planetary agencies shape their worlds and others, the dimensions along which their actions, forces, or effects are released on Earth, and how these *agency dimensions* relate to *planetary habitability*. Planetary habitability thus emerges as a pragmatic core concern within our political and theoretical argument. It conceptually expands on biological definitions of 'habitability' traditionally referring to the ability of an environment to support life—“the confluence of liquid water, free energy (chemical or solar), and the availability of the basic chemical ingredients of life (CHNOPS), bounded in parameter space by the physiological limits of life on Earth” (cf. Merino et al., 2019; Wong et al., 2022, p. 5).

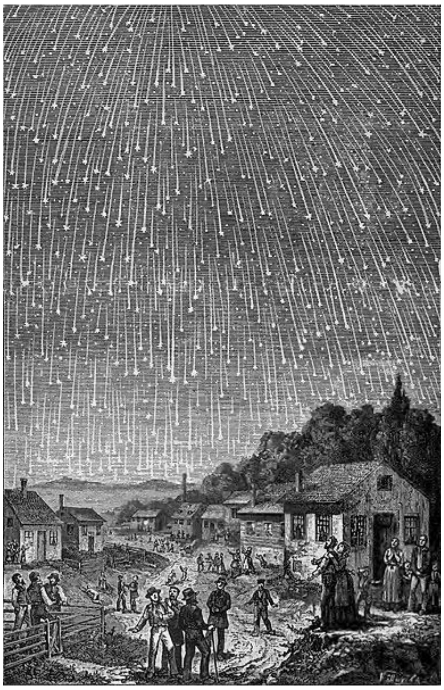
The concept of ‘life’ itself has, in recent decades, been progressively “unwound” due to proliferating biotechnologies, the bioengineering of synthetic forms of life, and the disciplinary rise of astrobiology as a field concerned with exploring the origins, existence, and potential diversity of ‘life as we know it’ and ‘life as we do not know it’ (*lyfe*) (Frank, 2024; Helmreich, 2016, p. 1; Roosth, 2017). Whereas conceptions of *life* are generally tied to the “specific disequilibria and classes of components of earthly life,” the conceptual term *lyfe* (pronounced: /loɪf/), introduced in 2020 by astrobiologists Stuart Bartlett and Michael L. Wong, “represents any hypothetical phenomenon in the universe that fulfills the fundamental processes of the living state, regardless of the disequilibria or components that it harnesses or uses” and thus might not conform to the biochemical or environmental constraints of Earth-based life (Bartlett & Wong, 2020, p. 6). To extend this conceptual boundary further, Wong and Bartlett recently proposed the term *genesity* “for a more general and expansive characterization of habitability” that distinguishes habitability as being about sustaining life under Earth-like conditions, while genesity is about fostering the emergence and continuous evolution of lyfe as a “new metric for generalized habitability” defined by the energetic driving force, the informational driving force, and the combinatorial diversity of components (Wong et al., 2022, pp. 1, 5).

While both expanded concepts of lyfe and genesity have their merit, especially beyond Earth (cf. Likavčan, 2024b), we maintain planetary habitability as the primary concern of a planetary politics *with* the Earth, building on Dipesh Chakrabarty’s distinction between sustainability and habitability as contrasting categories of the globe and the planet respectively (Chakrabarty, 2021, p. 81). Whereas “[s]ustainability is a deeply political idea in the Arendtian sense of the word *politics*” and tied to the human, “thus belongs firmly to the history of the global,” Chakrabarty writes, “[h]abitability does not reference humans,” instead it concerns “what makes a planet friendly to the continuous existence of complex life” (Chakrabarty, 2021, pp. 81, 83). Planetary habitability, as we conceive of it by expanding Chakrabarty’s focus on life further, thus emphasizes the broader co-constitutive entanglement of living, nonliving, and technological agents and agencies on Earth as they pertain to multispecies flourishing and thriving ecological systems. This shifts our focus unapologetically from human-centered affairs to a more inclusive co-habitability that concerns the planet's overall health and capacity to support diverse planetary agents and agencies. We use 'planetary habitability' in the following as a shorthand for the conceptual complexities unfurled above.

To realize more-than-human decision-making processes and solve conflicts of interest in service of planetary habitability, a nuanced understanding of the significant differences and overlaps between the dimensions on which the agencies of solar storms, mountain ranges, leaf cutter ants, and Large Language Models (LLMs) unfold is indispensable. We thus delineate these agency dimensions in the context of three—interrelated, interdependent, and at the edges challenging to separate—groups: nonliving, living, and technological planetary agency types, aiming to reflect where they lift weight off or bear weight on the Earth's vital cycles. If human societies find ways to work *with* more-than-human agents while staying with the trouble of our planetary condition, we may ultimately foster a more habitable world (cf. Bridle, 2022, p. 19; Haraway, 2016, pp. 99-103). For that aim, politics need to recognize what they have contributed all along. It is thus time to come to terms with how these planetary agencies practically play out.

1.3. Planetary Agency Types and Dimensions

On the night from November 12 to November 13, 1833, the public across North America experienced a stunning display of a veritable meteor storm, with estimates of 50,000 to 150,000 meteoroids per hour burning up in Earth's atmosphere over nine hours (Jenniskens & Butow, 1999). A letter sent to the *Daily Herald* by astronomer Denison Olmsted, in which the Yale professor appealed to the public to gather all information about the storm, became a catalyst for one of the first citizen science projects on the causes of the remarkable occurrence (Walker, 2020). Commonly assumed to be an atmospheric phenomenon at the time, Olmsted's systematic investigation revealed its cosmic origin. Known today as radiating from the constellation Leo, the so-called Leonids are debris particles left by the comet 55P/Tempel-Tuttle and an annual occurrence in November, albeit typically producing meteor showers with a mere 15-20 meteors per hour. Following a 33-year cycle tied to the comet's orbital period, reinforcing the debris trails along its orbit, the meteor storms setting the night's sky alight indelibly touched cultural imaginaries and histories across human societies. This extended from the mythological and cosmological to complex political affairs. In North America, pictographs inscribed by the Lakota on a bison hide suggest that peace and trade agreements between rivaling tribes were settled in "The Year the Star Fell" (Zhang, 2017, p. 11). At the same time, the stellar performance similarly left its mark on the abolitionist movement, with both enslavers and abolitionists referencing the night in 1833 as a pivotal moment for their respective causes (Rao, 2010). In Mesoamerica, Eastern Asia, and the Middle East, meteors have long played a leading role in timekeeping (e.g., Mayan calendar system), astronomical studies (e.g., Al-Sufi's *The Book of Fixed Stars*, published in 964), and influenced governance decisions as 'celestial mandates' in imperial China and Japan (Farris, 1992; Jiang, 2011).



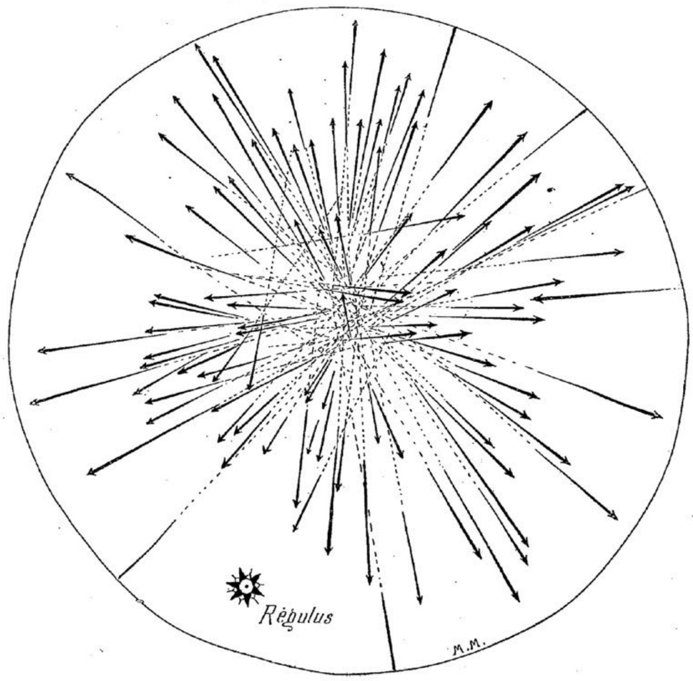


Figure 3 (left): Leonids meteor storm 1833, engraved by Adolf Vollmy, after a painting by Karl Jauslin. Public domain. **Figure 4** (right): Meteor shower observed from Venezuela by the German naturalist Alexander von Humboldt. *La Science Illustree* (May 1891): p.7. Public domain.

Known as *fundamental interactions*, the gravitational, electromagnetic, and strong and weak nuclear forces govern all interactions between particles in the universe (Dhiman, 2020; Rehm & Biggs, 2021). Two of these four fundamental forces in physics play a role in causing meteor showers on Earth. The Sun's gravitational force warping its surrounding spacetime continuum governs the movements of comets and asteroids in its proximity. When orbital meteoroid trails cross paths with the Earth, our planet's induced gravity pattern interferes with the gravitational agencies of the Sun and the meteoroids so that some of these cosmic particles are diverted and enter the Earth's upper atmosphere, where they encounter friction with atmospheric particles, causing rapid heating and ionization. This process involves electromagnetic force, as the ionized particles interact with Earth's magnetic field and atmospheric components, generating light and heat visible from Earth's surface as 'shooting stars' in the night sky (Capucci, 2019).

The fundamental forces and their interactional agencies drive innumerable physical gradients and parameters, each affecting and being affected by others to varying extents, with diverging degrees of what humans interpret as autonomy or freedom (Zaccagnino & Doglioni, 2022, p. 802ff.). These agencies manifest in energy flows that move particles, continuously shaping and transforming matter and biogeochemical processes (cf. Kleidon, 2016). By examining these planet-shaping processes across spatiotemporal scales, we gain a deeper understanding of planetary transformations and the cosmic foundations of our planetary condition. A brief survey thus follows of these fundamental forces and their planetary agencies on Earth, from which we develop a matrix of *planetary agency dimensions* that will serve as a theoretical framework to outline the scaffolds for a planetary politics.

Isaac Newton's law of universal gravity describes the weakest fundamental force as an attraction between masses in proportion to the distance between their centers (Newton, 1729). Thus, if an apple falls from a tree toward the ground, its small mass's gravitational force is congruent with the much greater one affecting the Earth. We are nevertheless "deceived by the fact that the Earth moves much less under the influences of these forces than the much lighter mass" because the ground's solid reaction force "exactly compensates the gravitational force" (Fratzl & Schäffner, 2021, p. 38). Albert Einstein's theory of general relativity defines gravity not as an external force but as a geometric property of mass and energy that affects space and time, bending the spacetime continuum into a curve (Einstein, 1915). In the presence of energy, any object with mass warps spacetime around itself. According to Einstein, the apple falls only along the curvature of space and time caused by masses without being subjected to an external force. It is the distortion of these dimensions that we perceive as gravity.

Meteor showers, therefore, emerge from an interplay of (inter-)planetary space-time bending, which creates trajectories for inertial systems, like the Leonids, to move force-free. We may conceive these fundamental interactions along the lines of a 'parliament of planets'—a solar version of the Latourian thought experiment that long predates human minds in its constitution and negotiates long-term arrangements on how planets "share the space around their sun" (Szersynski, 2021, pp. 209-210, 220). Without the fundamental agency of gravitational embedding, solar nebulae would be unable to form the rotating discs of hydrogen, helium, and metals that eventually accrete into clusters of matter (*planetesimals*), evolve into "planetary embryos" (Szersynski, 2021, p. 217), and ultimately form planets that then self-assemble and engender their gravitational agencies (cf. Basri & Brown, 2006). This *planetary becoming* discloses compartments of gravitational systems within a universe affecting its living and nonliving elements, including other planets. On Earth, the conditions for life to evolve were thus created. What distinguishes this parliament of planets is that "[p]lanets are never solely at the mercy of external forces; even the effect on them of a collision with another astronomical body or a change in brightness of their star will depend on how the planet has developed so far" (Kring, 2003; Szersynski, 2021, pp. 219-220). The Earth's interference and reproduction of gravitational agencies thus fundamentally shape our planetary condition. As a most recent addition to this interplanetary narrative, in 2022, NASA's DART mission made history by successfully altering the orbit of asteroid Dimorphos, inadvertently creating the first human-made meteor shower (Peña-Asensio et al., 2024), an incursion into the governmental realm of the parliament of planets.

The consequences of gravity's mediation of ecological or atmospheric cycles and energy transforming and channeling define planets and entire galaxies. On Earth, the planet's rotational patterns, geography, oceanic tides, and all aspects of human life are determined by it. Gravity thus demonstrates that humans, like any matter on Earth or the universe, are eternally subjected to the same agentic force. Like the apple, the Moon, the meteoroids, or the Sun, humans endlessly fall by the rules of gravity (Wapner, 2023). The fact that gravity traps humans on Earth and forces them to apply vast amounts of energy to counter its effects constitutes an indispensable constant of life, for in the absence of gravity, human bodies physiologically deteriorate (Ranganathan, 2016). Recent studies even suggest that human bodies may grow hypersensitive to gravitational forces on Earth due to stress factors such as weight gain or irregular sleeping patterns, potentially impacting their gastrointestinal health (Wapner, 2023). Fittingly, philosopher Yichao Lin recently argued through his examination of the historical transmissions and conceptual evolution of ‘planetarity’ from Western to East Asian contexts that “[a] precise definition of ‘planetarity’ has nothing to do with life and technology on spherical celestial bodies. Fundamentally, it must—and can only—be defined as the essential and singular gravitational dependency within a system of mass points. This represents the ultimate condition humanity can neither alter nor escape” (Lin, 2024).

Next to Earth’s distinct gravity, the Moon’s gravitational pull equally affects the ecological patterns on Earth, regularly causing the shift of oceanic tides, thus shaping human societies and their natural rhythms (Coughenour et al., 2009). The rising and falling sea levels and weather patterns provide the conditions for coastal ecosystems to thrive, form the basis of agricultural activities, render the geographies along coastlines and the related distribution of human settlements, are harvested as tidal energy by power mills, and determine human routines in fishing or sea travel. The other way around, Earth’s gravitational influence also “exerts forces on the Moon in the form of solid body tides that distort its shape” (Hille, 2015) and causes its gradual distancing from Earth by 3.8 cm annually, approximately the rate at which human fingernails grow (NASA, 2024).

Gravity is thus revealed as a fundamental force with planetary agency, affecting essential processes and interactions on Earth. It even played a pivotal role in the emergence of human life, which is linked to the impact of an asteroid on the Yucatán Peninsula 66 million years ago, widely acknowledged as the primary cause of the Cretaceous-Paleogene (K-Pg) extinction event that wiped out most dinosaurs and countless other organisms. This event marked the end of the Mesozoic era and indirectly set the scene for the emergence of mammals and, eventually, human life, illustrating the profound effects that mass extinction events can have on evolutionary processes (Longrich et al., 2016). As a result of the asteroid impact, immense amounts of sulfur and nitrogen compounds were released into the atmosphere, leading to global climate disruptions, including a dramatic cooling period caused by atmospheric particles blocking sunlight and resultant acid rain, affecting land and ocean ecosystems. Around the same time, enormous volcanic eruptions in what is now India (the Deccan Traps) emitted large quantities of greenhouse gases that contributed to prolonged global warming. Combined, the “one-two punch” effects of the Chicxulub asteroid and Deccan Traps volcanism created cascading ecological disruptions that reduced biological competition, indirectly benefiting the diversification and survival of early mammalian ancestors (Keller et al., 2020; Renne et al., 2013; Schulte et al., 2010).

On Earth, the fundamental forces are deeply entangled with our planetary condition. Next to gravity, the electromagnetic force contributes its life-enabling agency to generate Earth’s electromagnetic field, which shields the planet against solar winds and cosmic radiation (Wilde, 2019). This fundamental force composed of electricity and magnetism keeps electrons in orbit around the nuclei of atoms and gives rise to magnetic fields (Robinson & Neville, 2023). Neither navigation technologies, global communication networks, nor electric motors could have been developed without the attractive or repellent forces between positively charged atomic nuclei and negatively charged electrons (Stöhr & Siegmann, 2006, pp. 1-6). In 1859, a massive solar storm referred to as the Carrington Event caused intense auroras to dance about Earth’s magnetic field lines, causing likewise the collapse of telegraph networks and navigation technologies across North America and Europe (Baker, 2009; Gebhardt, 2020; Lanzerotti, 2017; Riley & Love, 2017). Sunspots tangling the star’s magnetic field lines between August 28-29 and September 1-2 caused solar flares and enormous coronal mass ejections that traveled through interplanetary space, hitting the Earth as severe geomagnetic storms. As the charged particles penetrated the Earth’s atmosphere, causing telegraph wires to catch fire under the massive voltage, the Sun’s ecology- and atmosphere-mediating agency materialized at an extreme, interfering precariously with human technology’s communication and information exchange.

Even outside severe occurrences like the Carrington event, the electromagnetic force crucially preconditions planetary habitability by serving atmospheric mediation. Without the planet’s electromagnetic field shielding its surface from the Sun’s radiating agency, life on Earth would be destroyed in most forms. The extreme force behind the Sun’s energy transforming and -transmitting agency emerges in its core, where temperatures reach fifteen million degrees Celsius, converting hydrogen gas into plasma—a process that interferes with the electromagnetic force holding the atoms together, thus separating the electrons from atomic nuclei (Sturrock, 1986a, 1986b, 1986c). These hydrogen nuclei are hence free to move at high speed, setting a series of nuclear fusion processes (proton-proton chain reactions) in motion that eventually fuse four hydrogen nuclei into one helium-4 nucleus and release an enormous amount of energy due to a mass defect in this fusion (ibid.). Convection flows and magnetic fields transmit this energy through the different layers of the Sun until it radiates out into space in the shape of photons, eventually reaching Earth (Anderl & Leggewie, 2024).

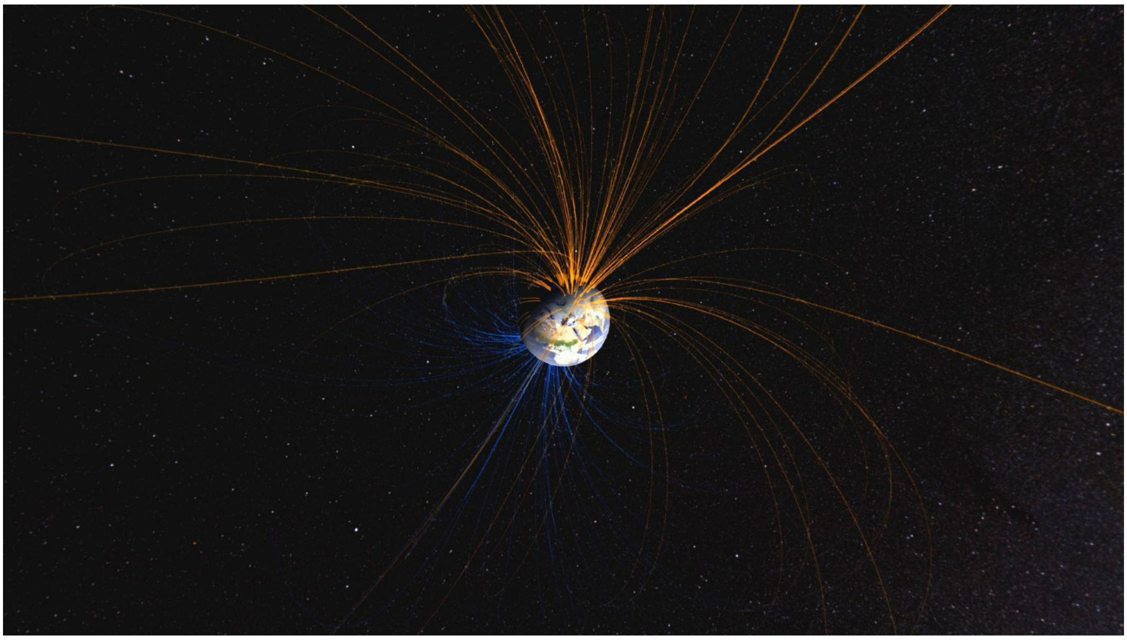


Figure 5 (above): Earth's magnetic field connects the North Pole (orange lines) with the South Pole (blue lines). **Figure 6** (below): Coronal mass ejection (CME) reaches Earth, the planet's magnetic field deflects solar particles. Still images from *Dynamic Earth: Exploring Earth's Climate Engine*. Created by NASA Goddard's Scientific Visualization Studio. (CC BY 2.0).

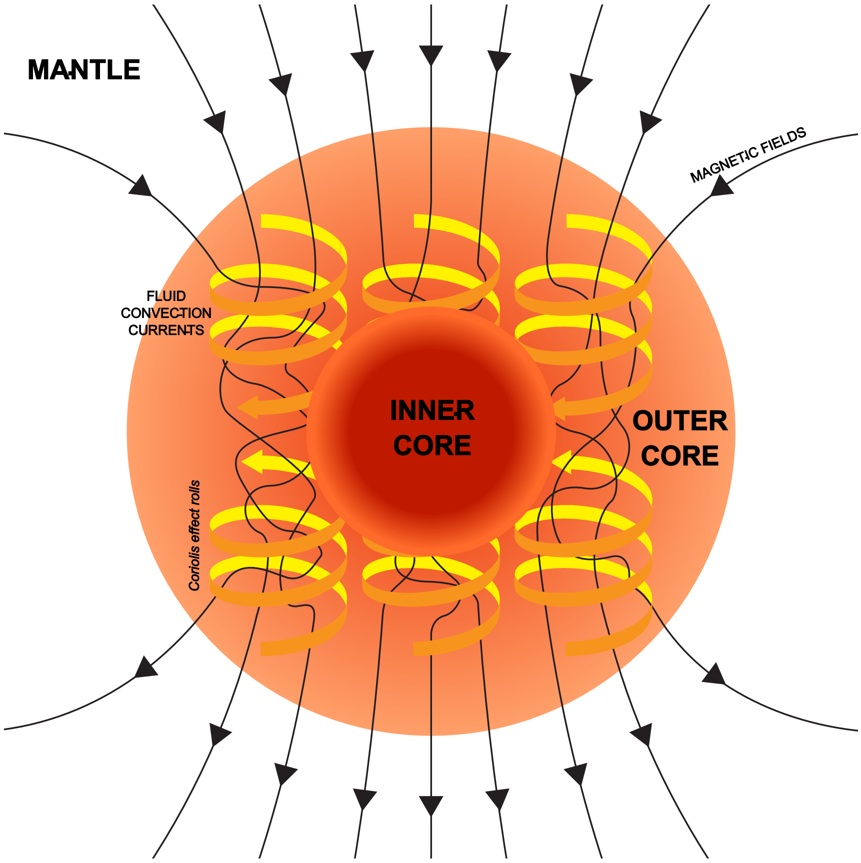


Figure 7: Diagram showing the relationship between the motion of conducting fluid (organized into rolls by the Coriolis force) and the magnetic field generated by the motion. Original work by the United States Geological Survey. Created by Andrew Z. Colvin. (CC BY 4.0).

Were it not for various processes and material constellations interacting and intra-acting within the planet's interior to engender its magnetic field, solar radiation would render life on Earth, as we know it, impossible (Stöhr & Siegmann, 2006, p. 62; Wilde, 2019, p. 204). Within the Earth's liquid iron outer core, convection flows transform mechanical into electromagnetic energy and—complemented by the Earth's rotation—drive a self-sustaining geo-dynamo: Energy is transported as warm matter rises and cools down so that it sinks back down, heats up, and rises again; the Coriolis force within the rotating core sustains these flows and causes electrically charged ions and electrons to move in spiraling motions (Wilde, 2019, p. 204). These flows continuously produce movements of charged particles from positive to negative poles and contribute to the electromagnetic field within the core extending from the Earth's interior into space (Christensen & Tilgner; Stöhr & Siegmann, 2006, p. 62; Wilde, 2019, p. 207). Within this field, two radiation belts, named after astrophysicist James Van Allen, intercept significant amounts of the charged particles that travel from solar winds or cosmic rays, keeping them from reaching the Earth's atmosphere (Li & Hudson, 2019).

The solar radiation that does reach Earth's surface does so as a continuous flow of solar energy, illuminating different parts of the planet as it rotates on its axis in its cosmic path orbiting the Sun. This energy crucially interacts with the planet's biosphere while also intra-acting with those entities and ecological processes at work on Earth (Crabtree & Lewis, 2007). Grappling with what this extraterrestrial agency means for our planetary condition relativizes anthropocentric and planetocentric perspectives, with potentially ground-breaking implications for human and more-than-human politics, as we will outline in this book. Neither solar radiation nor the fundamental forces operate in isolation but emerge in co-constitutive intra-actions. Thus, by relating these planetary agency dimensions, we illuminate the ways intra-active fundamental forces shape and precondition more concrete planetary agencies, providing and channeling energy for phenomena like solar storms, wildfires, ocean currents, jet streams, meteor showers, nuclear fallouts, and earthquakes. Recognizing the role of such agencies expands our political imagination, compelling us to consider governance systems that acknowledge the agency of nonhuman and more-than-human actors whose activities, shaped by cosmic and planetary dynamisms, profoundly influence our shared ecological futures on and with the Earth.

Accompanying its protective function, Earth's electromagnetic field also serves human and nonhuman life as a means of orientation regarding the agency dimensions of communication and information exchange. Interferences in these agencies may alter processes crucial for the survival of more-than-human beings or impede other dimensions in which their agencies unfold. This function is again tied to magnetic moments, the alignment of matter to its surrounding magnetic field. On the level of human technology, this effect was utilized by seafarers already in 1500 CE, who used lodestones—naturally occurring, magnetic minerals—as navigational tools to reliably mark directions due to their inherent ability to align with Earth's magnetic field (Stöhr & Siegmann, 2006, p. 3). Birds similarly utilize the electromagnetic field. On their seasonal migratory routes, they navigate by using landmarks, the position of the stars, and the Sun and by aligning their paths along electromagnetic field lines (Schulten et al., 1978). Research in quantum mechanics suggests that their internal compass relies on short-lived molecular fragments within their retinas, which are triggered into oscillation under the influence of the magnetic field (Hore, 2022). Within the cryptochrome protein, where this oscillation occurs, neurotransmitters are released, converting magnetic into biological information and passing it on to the avian brain, supporting its ability to ‘see’ magnetic field lines as if scanning a map (Wiltschko et al., 2021, pp. 6-7). Human-made electromagnetic fields (EMFs) radiating through air and oceans or space weather events causing electromagnetic field disruptions may thus disastrously interfere with the navigation of birds or bottom-dwelling marine species (Hutchison et al., 2020). Even more powerful interference of the electromagnetic field’s communicative and mediative agency dimensions is said to occur during geomagnetic field reversals, estimated to happen on Earth every 450,000 years (Cooper et al., 2021; Ratner, 2023).

On a smaller, if no less imperative scale, the strong and weak nuclear forces affect particles on a subatomic level. The strong nuclear force is essential for the stability of matter, as it binds protons and neutrons in atomic nuclei (Schmidt et al., 2020). The force is ‘strong’ in that it overcomes electromagnetic repulsion, which would lead to nucleus decay. It thus interferes with the energy-transmitting agency dimension in a manner that is preconditional to any form of co-creation performed by material planetary agencies. Post-1938, after humans first split an atom, reconfiguring mass into the release of nuclear energy, the ripple effects of this novel alliance with a fundamental force resulted in the momentous beginning of the Nuclear Age for human societies worldwide (cf. Märkl, 2019, pp. 7-40) while causing significant and irreversible damage to Earth (Schunck & Regnier, 2022). The infamous Manhattan Project, while forcing a turning point in World War II, likewise caused large-scale destruction and suffering by dropping atomic bombs on Hiroshima and Nagasaki in 1945, lastingly shaping the nature of geopolitical conflicts and warfare (Märkl, 2019, pp. 41-112). Next to the horrific effects of nuclear accidents on multiple generations of humans and other animals, the massive increase in nuclear testing during the Cold War era caused a doubling of carbon-14 in Earth’s atmosphere throughout the 1950s and 1960s (Malate, 2020). The ecology-mediating agency performed by carbon dioxide molecules absorbing infrared radiation and then re-emitting it back toward the planetary surface to keep it warm enough to harbor life (greenhouse effect) is thus augmented in a manner that leads to harmful interferences with Earth’s climate system (Liu et al., 2022).

Once the natural sciences better comprehend how these fundamental forces and their agencies function across the universe and on Earth, humans may form alliances with or learn to contain them (Slijepčević, 2023, p. 68). These alliances could significantly serve humanity if they are struck with adequate respect and care.

Withstanding the negative example of human hubris in attempting to control the strong nuclear force, research on the weak nuclear force has pushed the boundaries of scientific understanding and significantly helped humanity expand its agentic capacities (Cirigliano et al., 2022). This force transforms particles within atomic nuclei through radioactive decay processes and is fundamental to the evolution of stars and all earthly life. It facilitates the emergence of elements and allows the fusion of hydrogen nuclei into helium in the Sun, the origin of its life-giving glow (Robson, 2021; Sutton, 2023). Human efforts to understand and harness fundamental forces and their agencies have advanced our knowledge of the intra-actions among different forces. This pursuit may one day enable physicists to describe a so-called 'theory of everything' that unites gravity, electromagnetism, and the strong and weak nuclear forces to understand dark matter, dark energy, black holes, and other yet-to-be-resolved mysteries of physics (Pultarova, 2022).

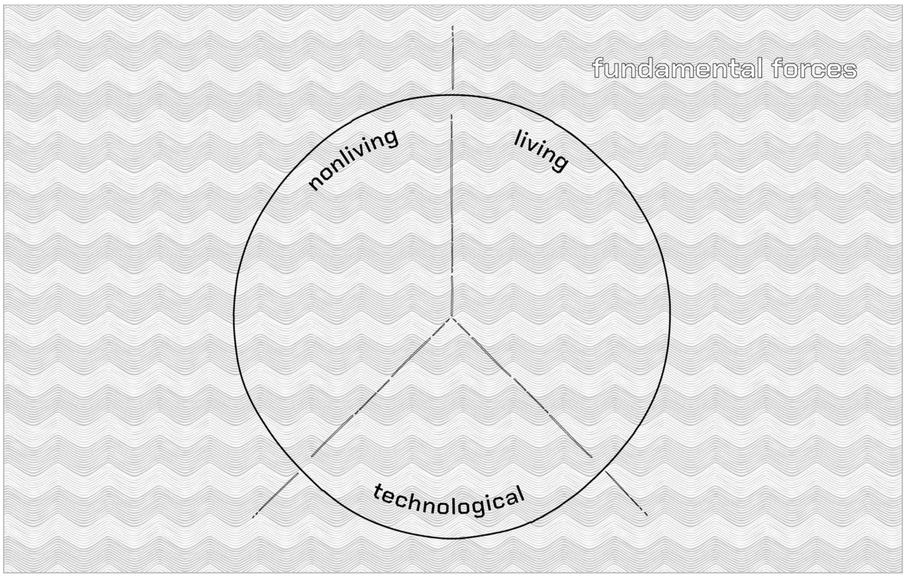


Figure 8: Planetary Agency Types (diagram). PLACEHOLDER (graphic design commissioned)

Building on this sweeping survey of fundamental forces and their diverse planetary interactions and intra-actions, we have chosen to classify and establish criteria that allow our analyses in this book to introduce a level of democratic order into an otherwise unwieldy swarm of planetary agents and agencies on Earth. Although recent technological and scientific developments have made conceptual boundaries between living and nonliving agents increasingly permeable, we unravel specific types of planetary agencies via three nonexclusive groups, with some representatives spanning more than one group (see also List of Terms).

The Agency of Nonlife (Chapter 2) preconditions cause-and-effect influence within broader planetary dynamics. Humanity has implicitly or explicitly allied with some of these nonliving agencies. Understanding how they shape our planetary condition is critical to building scaffolds for a planetary politics, where wildfires, earthquakes, or tropical cyclones reveal their distinct agency dynamics. In seeking a political 'voice' for the planet, the meaningful impacts of these nonliving entities may be most closely associated with that objective.

The Agency of Life (Chapter 3) contributes to planetary dynamics through complex, species-specific agencies that differ markedly from those of nonliving agents. By problematizing how natural scientists identify the origins of 'life,' we uncover the various dimensions in which living agencies shape our planetary condition. While nonliving planetary forces, like ocean currents or seismic vibrations, may transmit their planetary expressions more directly to human actors, the agencies of animals, plants, fungi, and protists emerge within the fabric of biotic communities, subtly shaping ecological and social structures. Given the closer alignment of humans with living agencies than nonliving ones, understanding living agents as more-than-human mediators offers critical political insights, especially when considering their roles in a broader framework of planetary politics.

The Agency of Technology (Chapter 4) serves as an artificial interface that can increasingly mitigate the semiotic and phenomenological fissures between humans and more-than-human agents. Whether defined by artificial intelligence (AI) or not, we discuss the agencies of robots, chatbots, and other technologies, problematizing whether and how we can interpret these potential planetary mediators as (a) giving a political voice to non-human and more-than-human agencies and (b) progressively engendering a political voice of their own.

The examples of various nonliving, living, and technological planetary agencies featured in the upcoming chapters are extensive. Still, we do not assert that they constitute a complete inventory—a Sisyphean task better suited to other formats. Rather, conceived as a general overview, our examples demonstrate the practical emergence of agency dimensions, their interrelationships, and their possible significance for a planetary politics. Each of these chapters sets off with a description of the respective type of planetary agency and how it differs from others. We then approach an understanding of their political voices by grasping the dimensions into which their agencies stretch. These reflections engender questions that must be raised before designing political frameworks that address them. Based on the analyses of nonhuman agencies—presented via instructive examples—we designate the following nine *planetary agency dimensions* that collectively capture shared characteristics relevant to examining our planetary condition. Throughout this book, we conceive these as a model framework of reference that allows us to inspect diverse planetary agencies. This helps grasp their complexities and approach the creation of political and social frameworks that may facilitate their full integration into a planetary politics based not on individual voices but on planetary agencies—agencies understood as dynamic intra- and interactive flows rather than discrete units. Not unlike the concept of 'flows' discussed in the legal and political frameworks of multispecies constitutionalism (cf.Blake & Webb, forthcoming; Celermajer et al., 2021; Chao & Celermajer, 2023; Fitz-Henry, 2022; Tschakert, 2022), we shift the focus from individual political entities to the maintenance and governance of relational agencies that sustain planetary habitability. Such a perspective reframes planetary politics as a practice of recognizing and aligning with these agencies as political shareholders, ensuring their efficacy, and mediating their impacts across human and non-human domains.

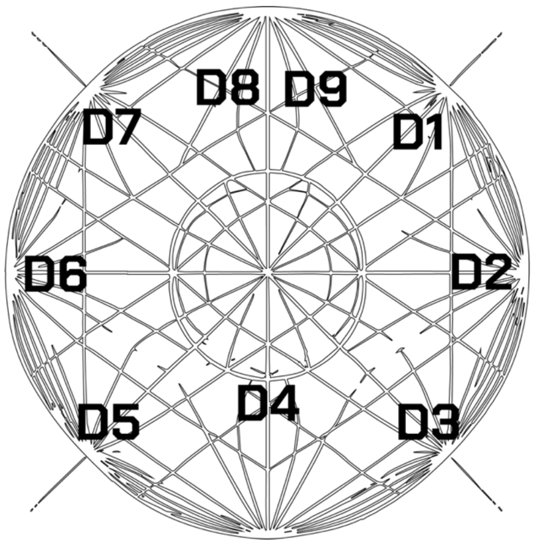


Figure 9: Planetary Agency Dimensions. PLACEHOLDER (graphic design commissioned)

1. Mediation of Ecological or Atmospheric Cycles (D1)

The first dimension on which planetary agents operate concerns their capacity to mediate and thereby (de)stabilize ecological or atmospheric systems and processes. We speak of ecological mediation when forests convert carbon into oxygen and thus contribute to regulating Earth's climate; wetlands filter pollutants from water and protect ecosystems, including their inhabitants, from excessive flooding; predatory animals keep populations of herbivores at quantities their respective ecosystems can sustain; or when water masses enrich the soil of flooded riverbanks thus assisting hydrophile plant and animal species to thrive. We speak of atmospheric mediation when extreme weather events release vast amounts of energy and materials to balance accumulated tensions or when cosmic forces steer energy and particles. Agencies reaching this dimension tend to lift a weight off the Earth as they perform essential functions for sustaining its critical biophysical and physicochemical cycles.

2. Energy Transformation and Channeling (D2)

Many planetary agents are capable of capturing, converting, and distributing energy for themselves or within ecosystems. When particles attract and repel one another, transporting electromagnetic charges, when microorganisms break down organic matter to release energy, or when plants convert solar into chemical energy through photosynthesis, we attribute their agencies to this second dimension. The energy transforming and -channeling processes set in motion by fundamental forces (of physics) precondition all planetary agencies. Without them, neither living nor technological entities could exist. Where life forms metabolize nutrients to generate physical energy, technological agents consume physical energy to perform mechanically. Notwithstanding their mode of energy transformation, all these agencies can be beneficial or harmful to the Earth, depending on the form and extent to which they are realized and implemented.

3. Material (Re)Cycling (D3)

We describe the capacity of planetary agents to (re)cycle and (re)distribute materials and essential nutrients within their respective environments as a third dimension of planetary agency. One example is soil fertilization through the metabolic byproducts of living agents. The ability of fungi or bacteria to decompose dead organic material is a crucial example. As with all agency dimensions, material (re)cycling depends not on one but several interrelated planetary agencies. For instance, some bacteria require oxygen to decompose matter, while other agents, like rivers, rely not only on fundamental physical processes but also on biological agents like fish or algae to transport seeds, nutrients, or sediment. Without material recycling enacted by these planetary agencies, vital resources like nitrogen, phosphorous, or carbon would quickly deplete. However, even when they provide the Earth's biochemical systems with essential components, agents and their material (re)cycling agencies may conversely threaten inhabitants of shared ecosystems. Suppose the quantity of decomposers in any given ecosystem rises to levels that cause system-wide disequilibrium. In that case, other life forms may be devastated and thus hindered from realizing their respective ecosystem-mediating agencies.

4. Communication and Information Exchange (D4)

The ability to share information and signals among and between planetary agents of the same or different kinds shapes the fourth agency dimension we describe. One thoroughly explored form of more-than-human communication is the waggle dance performed by honeybees to convey directions to food sources. The exchange of biochemical signals between trees via their root systems or allied mycorrhizal networks are among the numerous other examples that have long since demonstrated that human language is not the only semiotic system on Earth. Without communicative and information-transmitting agencies, species could not coordinate themselves and others, interact between the same or with other species, or adapt to their environments. For a planetary politics, comprehending these agencies is of the essence, as this may elucidate the conditions that nonhuman beings require to thrive. Even if individual communications do not directly decode diverse knowledges about the conditions that sustain a habitable planet, they provide insightful starting points. Technological planetary agencies have begun to play a crucial role in this regard, as researchers are increasingly applying digital technologies enhanced with AI-driven algorithms to deepen their understanding of nonhuman communicative systems and mediate between the two.

5. Co-Creation and Engineering (D5)

The capacity to build and shape physical and biological structures that influence ecosystems or facilitate the development and application of natural technologies plays into the fifth dimension. Numerous animal species function as more-than-human 'architects' or 'engineers,' creating habitats for themselves and others, learning to use tools, or developing and passing on survival strategies. Beavers construct dams that create wetland habitats, elephants dig water holes used by themselves and others, and corals form reefs where marine life forms settle and thrive. Nonliving planetary agents engage in co-creation and engineering, too. Bodies of water transform the cliffs or ice shields on their shores. Seismic forces cause Earth's tectonic plates to drift and landscapes to transform. At the same time, AI-enhanced oil drilling machines readily optimize extractive industries and their exploitation of ecosystems. Without co-creating and co-engineering planetary agencies, the Earth's physical and biological structures would not look like they do today. Perhaps they would not even provide habitable environments for any being. The way nonhuman agencies modify their environments, create habitats, and influence ecological dynamics offers essential lessons for a planetary politics that aims to navigate human engineering ventures for the sake of planetary habitability.

6. Problem-Solving and Adaptation (D6)

We classify agents and their agencies that humans may consider 'intelligent' as belonging to the dimension of problem-solving and adaptation. This entails their ability to address challenges, adapt to changing conditions through growth, behavior changes, or other solutions, harness energy, or defend against threats. Numerous nonhuman animals, for example, have developed elaborate hunting techniques, including octopuses, ravens, certain primates, and orcas. The optimized growth coordination of a slime mold reaching for its nearest food source offers another well-known example. Similarly, AI-enhanced technological agents demonstrate advanced problem-solving agencies, which often exceed those of humans. This agency dimension is critical to the survival and resilience in changing environments for *all* planetary agents, including humans. Depending on the physiological makeup of the agent in question, this dimension may take very different shapes. It may include the ability to innovate, find solutions to challenges, or be restricted to adjust behaviors, strategies, or surficial responses to altered circumstances. Concerning our planetary condition, it is of the utmost importance that humans learn from more-than-human problem-solving practices to expand their repertoire and explore how this dimension may be innovated to mitigate environmental disruption.

7. Cultural Practices and Aesthetic Contributions (D7)

Both living and technological planetary agents engage in numerous and diverse practices that influence social structures and ecological aesthetics. We may be biased, however, if we believe that human artists, authors, or musicians are the only ones thriving in that regard. Species-specific variants of cultural and aesthetic practices can likewise be widely observed amidst nonhuman agencies. The songs of birds, their nest building, and the courtship rituals of many other animals, as well as the choices pollinators make based on the characteristics of flower buds, the migratory paths larger animals choose while wandering through landscapes, and the shaping of landscapes by mountains and glaciers all touch upon this dimension. Cultural practices and aesthetic contributions play a significant role in the evolution of life. As evolutionary drivers, they are thus likewise crucial for the social structures among more-than-human collectives of respective agency types. As the recent advancement of chatbots and other LLMs (large language models) causes this dimension to grow ever more powerful and difficult to control, a planetary politics must address how these and other planetary agents shape sociocultural and ecological structures.

8. Population Support and Sustenance (D8)

Another vast and essential agency dimension concerns the ability of planetary agents to support and sustain populations, often realized by fulfilling essential ecological functions. As with regards to each one of the nine dimensions we describe, all types of agencies contribute here: Wind currents distribute the seedlings of trees, pollinators facilitate the reproduction of plants, legumes, and bacteria return nitrogen into the soil, and eusocial insects collaborate with decomposers to recycle nutrients back into the ecosystem to feed the rest of their population. Organisms that realize their roles within diverse food chains by being consumed by others equally contribute to this dimension. More-than-human populations could no longer thrive without population-supporting and sustaining planetary agencies, and ecosystems could no longer provide them with stable homes. Considering these agencies is crucial in a planetary politics, yet conflicts of interest between the diverse contributing agents cause significant challenges. How these may be managed, given planetary habitability, becomes the question.

9. Regeneration and Emergence (D9)

Lastly, planetary agents engage in various activities to recover from disturbances, maintain ecological balance, or drive the emergence of new life or habitats. As biological processes may cause the extinction or destruction of certain species, individuals, or ecosystems, they may drive the evolution, growth, or emergence of others or entirely new variants. The capacity of forests to regenerate after wildfires or of coral reefs to recover from bleaching events, for example, is frequently curtailed in our planetary condition, as the planetary agencies in question often face too many challenges at once. The ability to recover after injury or sickness in living planetary agents also addresses this dimension. We may understand the overarching aim of these agentic practices as resilience-building. Some animal species engage in activities that resemble human medicinal practices to heal themselves and others. Fostering this agency dimension as effectively as possible is of the essence in a planetary politics. Without the capacity to withstand disturbances and recover from environmental stressors, the Earth's ecosystems cannot regenerate. Anthropogenic planetary distress could no longer be mitigated once planetary boundaries are crossed. To grapple with our planetary condition, it is crucial that a planetary politics not only supports but also learns from more-than-human agencies of regeneration and emergence. This way, strategies in fighting biodiversity loss or wildfires could reflect this knowledge and become more effective.

Coda. Specific planetary agents can restrict, confine, or enhance these dimensions. Human associations with planetary agencies, or their interference with them, for instance, have caused the emergence of various anthropogenic planetary agencies, including nuclear waste and climate change, with massive impacts on various planetary agency dimensions. Each agential interaction thus either lifts weight off or bears weight on the Earth, as illustrated by human-created flood interventions, population control measures, or geoengineering. Planetary agencies are in a continuous state of interaction and intra-action, where the efficiency of one agency often comes at the expense of another's potential realization. How these complex relations may be addressed and integrated into planetary governance becomes one of the compelling challenges and dynamic openings for a planetary politics.

The analytical scheme of nine planetary agency dimensions points to the methodological foundation of this book. It contributes to an eponymous transdisciplinary research program in the humanities and social sciences that takes the "Planet as Method." This means approaching the planet not as merely an abstract background or passive setting but as an active methodological agent, shaping knowledge production, cultural practices, and social organization. By invoking the planet as an intrinsic methodological tool, we aim to rethink foundational questions about agency in ways that are attuned to the profound interconnectedness of planetary systems. This also means fostering an awareness of the *astronomical planet* as firmly embedded within a cosmic ecology governed by fundamental forces and solar-terrestrial, interplanetary, and astronomical relations (cf. Likavčan, 2024a). Adopting the planet as a method thus requires both conceptual and practical shifts.

Conceptually, it demands a departure from frameworks that centralize human experience and instead encourages perspectives that recognize the agency of a wide array of planetary forces and entities. Practically, this approach encourages responsive and adaptive methods that can effectively interact with the planet's trans-scalar and multi-dimensional dynamics. Such an approach advocates for developing tools and frameworks designed to capture the complexities of planetary processes and their connections to social, ecological, and geopolitical systems. In this sense, "Planet as Method" functions as both a critique and an expansion of traditional research methods, bridging disciplines and scales to offer insights into the global, local, and more-than-human dimensions of planetary coexistence. It invites scholars to engage with planetary forces as active collaborators in knowledge production, fostering a relational understanding emphasizing co-adaptation and shared vulnerability. Through this, the book seeks to lay the groundwork for methodologies that analyze planetary aspects and actively participate in shaping a planetary politics that may govern more-than-humans toward a habitable (future) Earth. In order to analyze our planetary condition and trace its political consequences, we review, compile, and discuss various sources in the form of qualitative meta-analyses, using examples to illustrate and validate the plausibility of yet relatively unfamiliar political forces (Dryzek, 2024; Levitt, 2018).